# Integrating Symbolic and Neural Mechanisms for Adversarially Robust Hyperdimensional Computing


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## Abstract

Neuro-symbolic models may improve robustness by combining learned representations with structured composition, but their behavior under adversarial perturbation remains underexplored. We study a hybrid pipeline that fuses ViT features with classical descriptors through Hyperdimensional Computing (HDC). Across CIFAR10 (Subset), COIL100, and ETH80 under FGSM and Genetic Attack, ViT + Classical HDC degrades more gracefully than Pure HDC and ViT + HDC baselines. It achieves higher normalized AURC, lower attack-time decision margins, and larger gains from partial adversarial retraining. These results suggest that classical-neural fusion within HDC is a promising direction for robustness under the evaluated threat models.



## CCS Concepts

• **Computing methodologies** → **Machine learning**; **Computer vision**; • **Security and privacy** → *Software and application security*.

## Keywords

hyperdimensional computing, neurosymbolic AI, adversarial robustness, robust machine learning


## 1 Introduction

The rapid rise of transformer architectures and large language models (LLMs) has renewed interest in combining the two dominant paradigms of artificial intelligence (AI): Symbolic AI and Neural Networks. Symbolic methods support explicit reasoning, interpretability, and prior-knowledge injection, but struggle with high-dimensional raw inputs and continual adaptation. Neural models learn effectively from pixels or tokens and scale well, but are often opaque and vulnerable to adversarial perturbations. Neuro-symbolic AI seeks to combine these complementary strengths (Table 1) [19].

| Capability / Criterion | Symbolic AI | Neural Networks |
|---|---|---|
| Robust against adversarial attacks | ✓ | ✗ |
| Interpretable and explainable | ✓ | ✗ |
| Efficient in computational resources and energy | ✓ | ✗ |
| Generalization from raw data and adaptability | ✗ | ✓ |
| Handling large-scale or unstructured data | ✗ | ✓ |
| Flexibility across tasks | ✗ | ✓ |

**Table 1: Symbolic and neural methods provide complementary strengths that neuro-symbolic AI aims to combine.**

As AI systems move into safety-critical settings, adversarial robustness remains a central concern. Small, carefully crafted perturbations can induce incorrect high-confidence predictions in otherwise accurate classifiers (Figure 1) [7], and modern robustness evaluation extends beyond early one-step attacks to stronger iterative, adaptive, and black-box threat models [6, 12].

Neuro-symbolic AI is a natural setting for this question, but adversarial behavior in hybrid symbolic-neural pipelines remains less understood than in conventional deep models [6, 12, 19]. Symbolic structure alone does not guarantee robustness: weaknesses in learned encoders, fusion mechanisms, or downstream reasoning can still compromise the full pipeline. We therefore examine how neuro-symbolic design choices affect adversarial degradation, decision behavior, and recoverability under attack.

Hyperdimensional Computing (HDC) provides a useful substrate for this study. HDC represents information with high-dimensional distributed vectors and operations such as binding, bundling, and

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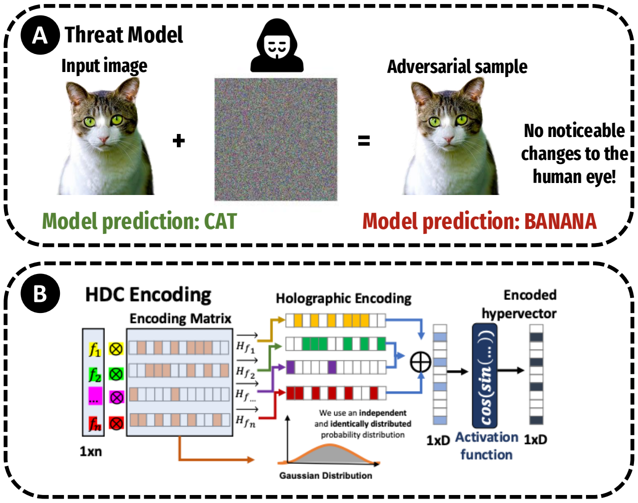


**Figure 1: Small perturbations can induce confident misclassification in deep models; HDC-based symbolic structure aims to improve robustness.**

permutation, enabling structured composition while remaining compatible with learned representations [9, 10]. Building on this perspective, we study hybrid symbolic-neural models that combine deep feature extractors with HDC-based composition. By integrating texture, shape, and color descriptors with HDC operations, the framework emphasizes richer relational structure beyond purely local pixel-level cues.

Our contributions are threefold. First, we develop unified pipelines that combine classical descriptors, neural feature extractors, and HDC-based composition into a single similarity-based inference framework. Second, we study robustness-oriented representations that emphasize structural and relational cues—texture, shape, and color—rather than relying exclusively on local pixel information that adversarial perturbations most directly corrupt. Third, we evaluate degradation behavior, decision-margin stability, and retraining recoverability under FGSM and Genetic Attack across symbolic, neural, and hybrid variants on three object-centric datasets. These complementary views of robustness consistently support the same conclusion: classical-neural fusion within HDC degrades more gracefully than purely symbolic or purely neural-HDC baselines under the evaluated threat models, and responds more strongly to partial adversarial retraining.

## 2 Related Work

Adversarial machine learning has shown that deep networks are vulnerable to small perturbations that induce incorrect predictions. Early work introduced FGSM [7, 18], followed by stronger iterative and black-box attacks such as PGD, DeepFool, and Carlini–Wagner [3, 4, 6, 11–13, 15]. Common defenses—adversarial training, distillation, input transformation, and randomized smoothing—improve robustness under specific assumptions but often at a cost in accuracy or generality [1, 3, 5, 7, 12, 14].

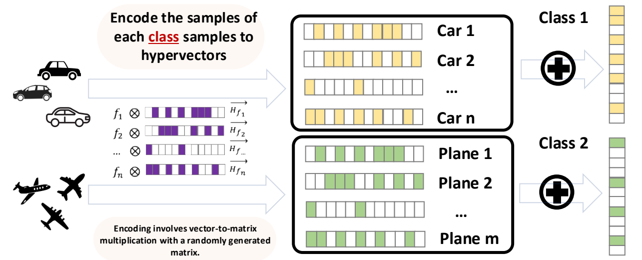


**Figure 2: Sample encoding and class-level bundling in HDC. Inputs are projected into hypervectors and aggregated into class prototypes.**

In contrast, adversarial robustness in neuro-symbolic systems remains less understood. Neuro-symbolic AI combines learned representations with structured symbolic mechanisms to improve interpretability and compositionality [19], but hybrid pipelines do not inherit robustness automatically because learned encoders, fusion stages, and downstream reasoning can still fail under attack. HDC is a natural substrate for such models because high-dimensional distributed representations support algebraic composition through binding, bundling, and permutation [9, 10], and prior work has explored HDC in efficient, brain-inspired, and hardware-aware learning settings [2, 8, 16, 17, 20–23]. However, adversarial robustness in pipelines that jointly combine classical descriptors, neural features, and HDC-based composition remains comparatively underexplored. Our work addresses this gap by comparing symbolic, neural-HDC, and classical-neural HDC variants under FGSM and Genetic Attack.

## 3 Hyperdimensional Computing

Hyperdimensional Computing (HDC) represents information using high-dimensional distributed vectors, or hypervectors, together with algebraic operations such as binding, bundling, and permutation [9, 10]. In this work, HDC is the composition layer that integrates heterogeneous feature sources, including classical descriptors and learned neural features, into class-level prototypes.

Given an input feature vector $\mathbf{x}$, encoding maps it to a hypervector $\boldsymbol{\phi} = \phi(\mathbf{x}) \in \mathbb{R}^D$. Encoded samples are then aggregated into class prototypes by bundling, while binding and permutation are used when structured composition or role information is needed. For each class $c$, the prototype is formed by averaging encoded samples,

$$\mathbf{C}_c = \frac{1}{N_c} \sum_{i=1}^{N_c} \mathbf{h}_i,$$

where $N_c$ is the number of training samples in class $c$.

At inference time, a test hypervector is assigned to the most similar class prototype using cosine similarity:

$$\hat{y} = \arg\max_c \frac{\mathbf{h}_{\text{new}} \cdot \mathbf{C}_c}{\|\mathbf{h}_{\text{new}}\| \, \|\mathbf{C}_c\|}.$$

Thus, HDC is used here not as an isolated contribution, but as the symbolic-neural composition mechanism through which we study robustness under adversarial perturbation.

# 4 Framework

Figure 3 illustrates the proposed hybrid symbolic-neural pipeline. The framework combines three components: (1) a classical descriptor branch that captures texture-, shape-, and color-related cues, (2) a neural branch that provides learned semantic representations, and (3) an HDC composition layer that integrates these heterogeneous features into class-specific prototypes for similarity-based inference. Classical descriptors such as LBP, HOG, and PHOG provide structural cues that are less directly tied to globally optimized pixel activations, while the neural branch contributes semantic abstraction. HDC serves as the shared composition space, allowing the final prototype-based decision to draw on both structural stability and semantic richness.

## 4.1 Model Variants and Neural Feature Branch

To study robustness across symbolic, neural, and hybrid regimes, we evaluate three model variants:

- **HDC Classifier**: a symbolic baseline that encodes extracted features into a high-dimensional space and classifies via similarity to class prototypes.
- **ResNet18 + HDC**: a hybrid symbolic-neural pipeline in which a pretrained ResNet18 provides learned visual features and HDC serves as the composition and classification layer.
- **ViT + HDC**: a transformer-based hybrid pipeline in which ViT produces global, attention-driven features that are then encoded and classified through HDC.

These variants allow us to compare purely symbolic and hybrid symbolic-neural formulations under the same HDC-based decision mechanism. In the hybrid models, the neural branch provides learned semantic representations, while the HDC head performs feature composition and prototype-based inference.

## 4.2 Classical Descriptor Branch

The classical branch is motivated by interpretable descriptors that preserve structural and relational information from the input image. We consider classical cues related to texture, shape, and color, since these are prominent sources of visual information that may remain informative even when learned embeddings degrade under perturbation. Figure 4 illustrates representative classical cue responses under clean inputs and Gaussian noise, confirming that Local Binary Pattern (LBP) texture maps and Sobel edge maps remain structurally coherent despite visible pixel-level corruption. Figure 5 further shows that these responses remain stable under adversarial perturbations, including FGSM and Genetic Attack.

For the reported quantitative experiments, the instantiated classical branch uses LBP, HOG, and PHOG as a compact descriptor set emphasizing texture and edge/shape structure. This choice reflects the final evaluation configuration used in the ViT + Classical HDC pipeline.

## 4.3 HDC Encoding and Feature Fusion

The extracted classical descriptors and learned neural features are mapped into HDC space and composed into a unified symbolic-neural representation. Figure 2 shows the encoding process used to convert feature vectors into class-level hypervector representations.

For the reported experiments, let $\mathbf{H}_{\text{neural}}$, $\mathbf{H}_{\text{LBP}}$, $\mathbf{H}_{\text{HOG}}$, and $\mathbf{H}_{\text{PHOG}}$ denote the encoded hypervectors from the neural and classical branches. These are combined to produce the final representation:

$$\mathbf{H}_{\text{final}} = \mathbf{H}_{\text{neural}} \oplus \mathbf{H}_{\text{LBP}} \oplus \mathbf{H}_{\text{HOG}} \oplus \mathbf{H}_{\text{PHOG}}, \tag{1}$$

where $\oplus$ denotes the HDC composition operation used in the implementation. This design allows heterogeneous sources of information to be integrated within a shared representational space rather than merged through simple concatenation alone.

## 4.4 Prototype Formation and Inference

Encoded training examples are aggregated into class-specific prototypes, as described in Section 3. During inference, each test example is mapped into the same HDC space and assigned by similarity-based retrieval. The classical branch contributes structural cues, the neural branch contributes semantic information, and the HDC layer composes both before prototype comparison. Our hypothesis is that this shared classical-neural representation degrades more gracefully under adversarial perturbation than purely symbolic or purely neural-HDC alternatives. Because the final decision aggregates structural and semantic evidence, it is less dependent on any single corrupted feature source. The dimension sweep in Section 5.4 further checks that the observed gains are not artifacts of one hypervector size.

## 4.5 Threat Model

We evaluate the framework under two primary adversarial settings: FGSM and Genetic Attack. Together, these provide complementary white-box and gray-box views of degradation behavior: FGSM probes robustness when gradient information is available, while Genetic Attack evaluates gradient-free perturbations under partial observability.

**(1) FGSM (Fast Gradient Sign Method)** is a first-order white-box attack that perturbs inputs in the direction of the loss gradient:

$$x' = x + \epsilon \cdot \text{sign}(\nabla_x \mathcal{L}(x, y)), \tag{2}$$

where $x$ is the input, $y$ is the label, and $\epsilon$ is the perturbation budget.

**(2) Genetic Attack** models a gray-box setting in which the attacker observes model outputs such as logits or loss values, but not gradients. Adversarial examples are evolved through mutation and selection guided by a fitness score, providing a gradient-free attack mechanism for partially observable systems.

We also consider random Gaussian noise only as a sanity-check baseline for noise tolerance, not as a primary adversarial result. The main discussion therefore focuses on FGSM and Genetic Attack. Extending the analysis to stronger iterative and adaptive attacks is a natural next step.

# 5 Evaluation

## 5.1 Experimental Setup

We evaluate the proposed framework on three object-centric datasets: CIFAR10 (Subset), COIL100, and ETH80. CIFAR10 (Subset) denotes an approximately balanced subset with 5,000 training images and 1,000 test images, and the same fixed split is used across all models. Following the scoped threat model introduced earlier, we report results under two primary attack families: FGSM and a gray-box

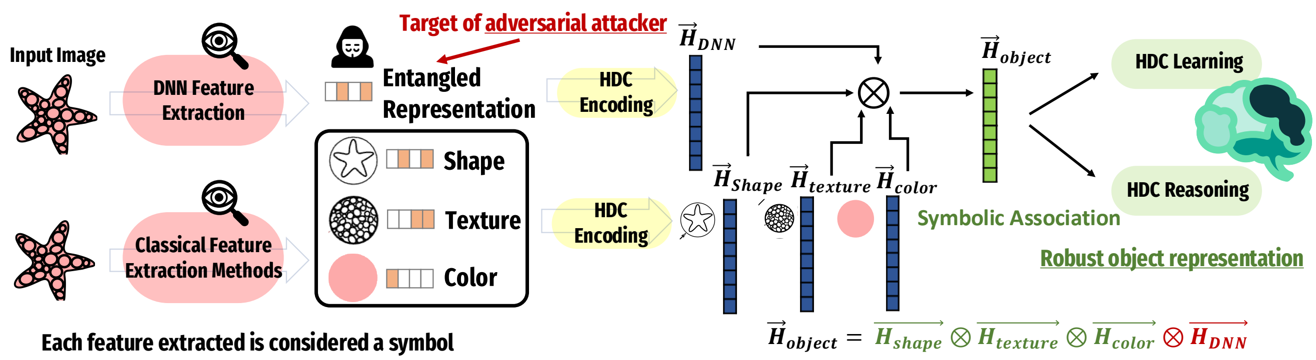


**Figure 3: Neurosymbolic pipeline. Neural features and classical shape-, texture-, and color-related cues are encoded and composed in HDC for similarity-based inference under adversarial inputs.**

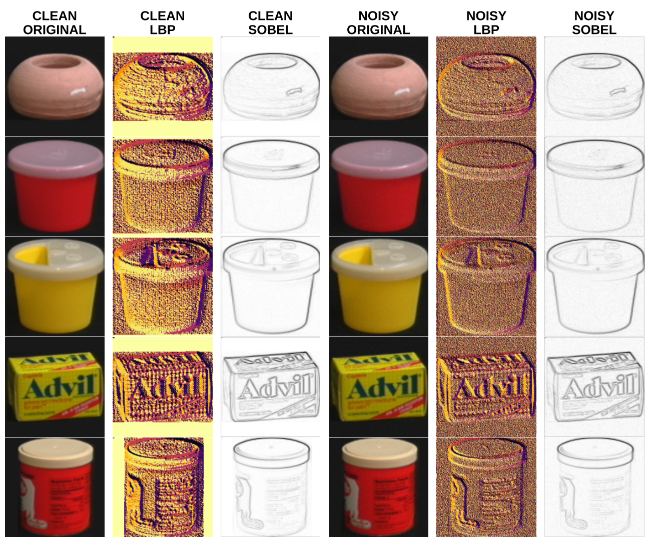


**Figure 4: Classical cue responses under Clean and Gaussian Noise ($\sigma = 0.015$) for representative COIL100 objects. LBP texture maps and Sobel edge maps remain structurally coherent despite visible pixel-level corruption, motivating classical descriptors as stable complements to learned features.**

Genetic Algorithm (GA). FGSM is used as a first-order white-box baseline, while GA is evaluated at larger perturbation budgets because it is weaker in practice under the considered setup:

$$\epsilon_{\text{FGSM}} \in \{0.0005, 0.001, 0.002, 0.004, 0.006, 0.008, 0.01, 0.02\},$$
$$\epsilon_{\text{GA}} \in \{0.01, 0.02, 0.05, 0.1, 0.2\}.$$

All HDC-based models use OnlineHD with hypervector dimension $D \in \{1024, 4096, 10000\}$. We compare three groups of models: (i) neural baselines (ViT Base and ResNet18), (ii) neural HDC hybrids (Pure HDC, ResNet + HDC, and ViT + HDC), and (iii) the proposed hybrid pipeline, denoted ViT + Classical HDC, which augments transformer features with a compact set of classical descriptors through HDC-based fusion. Because all symbolic and hybrid variants share the same HDC-based decision mechanism, these comparisons isolate the effect of representational content rather than changes in the downstream classifier itself.

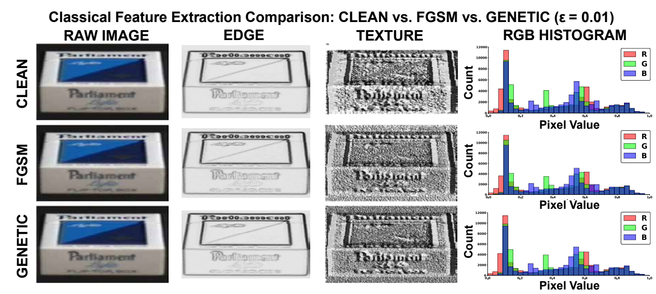


**Figure 5: Classical cue responses under Clean, FGSM, and Genetic conditions. Edge-, texture-, and color-related responses remain visually stable across adversarial perturbations, motivating the use of classical cues alongside learned features.**

For the reported quantitative experiments, the classical branch is instantiated with LBP, HOG, and PHOG descriptors. We study two variants: **Norm**, where descriptors are extracted after normalization, and **Denorm**, where descriptors are computed from pixel-denormalized input.

To keep the presentation compact while still covering the full experimental space, different figures emphasize different aspects of the evaluation. FGSM robustness heatmaps are shown for CIFAR10 (Subset) and ETH80 because these two datasets span a meaningful range of visual complexity: CIFAR10 (Subset) is a multi-class natural-image benchmark with substantial intra-class variation, while ETH80 is a lower-complexity object-centric dataset where structural cues are more salient, making the separation between methods more visible at larger perturbation budgets. Partial adversarial retraining is reported for COIL100 under FGSM and for CIFAR10 (Subset) under GA because these are the cases where the contrast between ViT + Classical HDC and ViT + HDC is most clearly visible after fine-tuning; Table 2 then aggregates the post-retraining accuracy gap across all datasets so that no individual case is over-emphasized. All reported comparisons use the same attack budgets, model variants, and HDC dimensions throughout; all experiments use fixed splits and shared preprocessing, with

ViT-family models operating at $224 \times 224$ resolution and the classical/HDC branch on resized $64 \times 64$ inputs. Descriptor extraction for the Norm variant is applied after per-image normalization, while Denorm computes descriptors from pixel-denormalized input; both are included because normalization choices interact with descriptor sensitivity under perturbation and together bound the range of classical branch behavior across the reported experiments.

## 5.2 Robustness Without Retraining

To summarize robustness across the evaluated perturbation range, Figure 6 reports the normalized Area Under the Robustness Curve (AURC) at $D = 10000$ on CIFAR10 (Subset). Normalized AURC captures average adversarial accuracy across the tested epsilon values, where higher values indicate slower degradation under attack. We use it here as a compact aggregate metric because it summarizes the full perturbation sweep in a single value and avoids over-interpreting any one epsilon setting in isolation.

The proposed ViT + Classical HDC model achieves the highest AURC, indicating that its accuracy remains higher on average across the evaluated perturbation range than the neural HDC and purely symbolic baselines. This provides a compact summary of the same trend examined in more detail later through budget- and dimension-wise FGSM heatmaps.

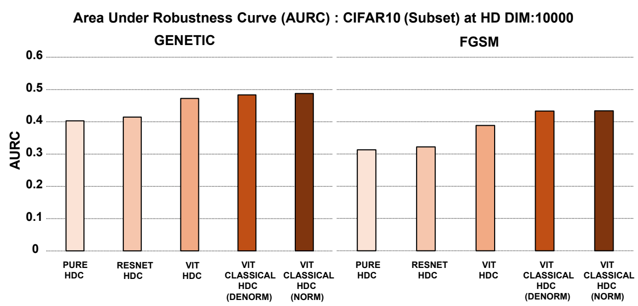


**Figure 6: Normalized Area Under the Robustness Curve (AURC) on CIFAR10 (Subset) at hypervector dimension $D = 10000$. Higher values indicate slower accuracy degradation across the evaluated perturbation sweep. ViT + Classical HDC achieves the highest AURC among the evaluated models.**

## 5.3 Decision Margin Behavior

We further analyze decision margins to examine how model confidence changes under attack. For each sample, the margin is defined as the difference between the highest predicted class score and the score assigned to the ground-truth class, so larger values indicate stronger separation in favor of an incorrect prediction and correspond to more overconfident failures under attack. Under GA perturbations (Figure 7), the margin distributions of all models shift upward, but the two ViT + Classical HDC variants remain both lower and tighter than ViT + HDC. On CIFAR10 (Subset) at $\epsilon = 0.1$, both Norm and Denorm produce visibly smaller margins than the neural HDC baseline, indicating less severe confidence inflation under attack. The same pattern persists on ETH80 at $\epsilon = 0.2$, with Denorm showing the most stable distribution among the compared models. These results complement the AURC summary in Figure 6 by showing that the hybrid classical-neural representation not only preserves accuracy more effectively, but also tends to fail less aggressively when errors do occur.

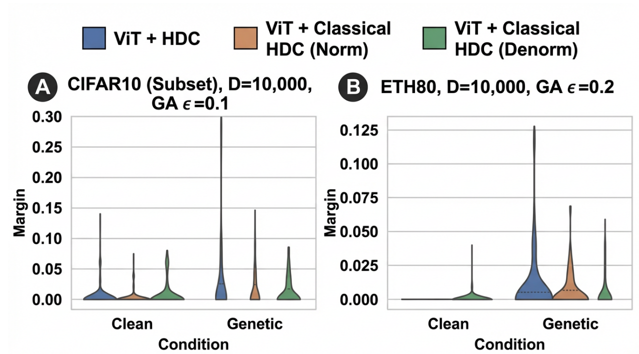


**Figure 7: Decision margin distributions under Genetic Algorithm attack. (A) CIFAR10 (Subset) at $D = 10000$ and epsilon $0.1$. (B) ETH80 at $D = 10000$ and epsilon $0.2$. Lower and tighter distributions indicate less overconfident failure under attack. Across both datasets, the classical fusion variants remain more stable than ViT + HDC.**

## 5.4 Effect of Budget and Hypervector Dimension

Figure 8 shows FGSM accuracy as a function of perturbation budget $\epsilon$ and hypervector dimension $D$ for (A) CIFAR10 (Subset) and (B) ETH80. On CIFAR10 (Subset), clean performance follows a consistent ordering,

Pure HDC < ResNet + HDC < ViT + HDC < ViT + Classical HDC.

This ordering is largely preserved as $\epsilon$ increases. Although all models lose accuracy under stronger perturbations, the hybrid classical-neural pipeline shows the most gradual degradation across the FGSM sweep, indicating that its advantage is not limited to the clean regime.

A similar trend appears on ETH80, where the separation between methods becomes more visible at larger perturbation budgets. In this setting, ViT + Classical HDC retains higher accuracy deeper into the FGSM range, while Pure HDC remains comparatively competitive only at smaller budgets, consistent with the lower visual complexity of the dataset. Across both datasets, increasing $D$ generally improves performance for all HDC-based models, but dimensionality alone does not explain the gains of the proposed model. The most consistent improvement appears when classical descriptors are fused with neural features.

## 5.5 Partial Adversarial Retraining

We finally evaluate partial adversarial retraining using a fixed 40/60 fine-tune/evaluation split under adversarial augmentation. Rather than showing retraining curves for every dataset and attack combination, Figure 9 highlights two representative cases where the contrast between ViT + Classical HDC and ViT + HDC is most

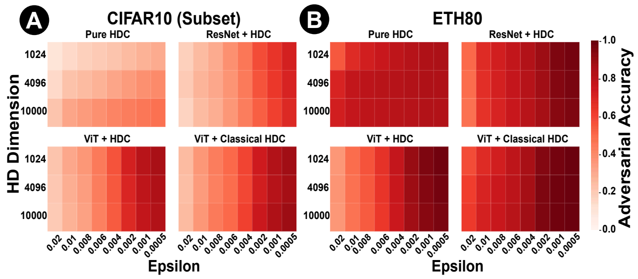


**Figure 8: FGSM accuracy heatmaps across perturbation budget $\epsilon$ and hypervector dimension $D$ for (A) CIFAR10 (Subset) and (B) ETH80. Rows correspond to $D = 1024, 4096, 10000$, and columns correspond to the evaluated epsilon values. Across both datasets, ViT + Classical HDC maintains the most stable robustness profile.**

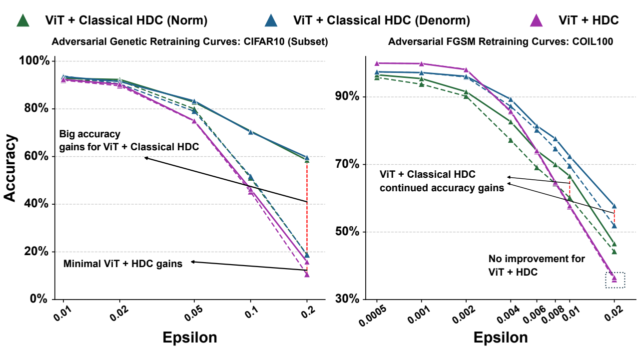


**Figure 9: Representative partial adversarial retraining results on CIFAR10 (Subset) under Genetic Algorithm and COIL100 under FGSM. Across both cases, ViT + Classical HDC shows clearer and more consistent gains from retraining than ViT + HDC.**

visible: CIFAR10 (Subset) under Genetic Attack and COIL100 under FGSM. In both settings, classical descriptor fusion yields clearer and more sustained gains across the perturbation range, whereas ViT + HDC improves only modestly and can stagnate or degrade at larger budgets. Table 2 confirms that this is not an isolated visual effect: averaged across datasets at $D = 10000$, the post-retraining accuracy gap grows from +6.57 to +10.94 percentage points under FGSM as $\epsilon$ increases from 0.01 to 0.02, and from +17.07 to +26.27 percentage points under Genetic Attack as $\epsilon$ increases from 0.10 to 0.20.

### 5.6 Summary

Across the evaluated datasets and attack settings, ViT + Classical HDC shows slower accuracy degradation, tighter decision margins, and larger retraining gains than ViT + HDC and pure HDC, supporting classical-neural fusion within HDC under the evaluated threat models.

**Table 2: Average post-retraining accuracy gap between ViT + Classical HDC and ViT + HDC at $D = 10000$, measured in percentage points and averaged across datasets.**

| Attack | Epsilon | Average Gap (pp) |
|---|---|---|
| FGSM | 0.01 | +6.57 |
| FGSM | 0.02 | +10.94 |
| Genetic | 0.10 | +17.07 |
| Genetic | 0.20 | +26.27 |

## 6 Conclusion

We presented a neurosymbolic framework that fuses classical descriptors with transformer features through an OnlineHD head and evaluated it across three object-centric datasets under FGSM and Genetic Attack. The core design premise is that descriptors encoding texture, edge, and color structure through LBP, HOG, and PHOG are less easily disrupted by the small-magnitude pixel perturbations considered here than learned embeddings alone, and that fusing them with ViT features inside a shared HDC space allows prototype-based inference to draw on both structural stability and semantic richness simultaneously. Because all symbolic, neural, and hybrid variants are compared within the same HDC-style inference setting, the observed gains are most naturally attributed to representational content rather than to a different downstream decision rule.

ViT + Classical HDC consistently showed the strongest robustness behavior among the compared configurations and was the only model to convert partial adversarial retraining into clear, sustained gains over ViT + HDC (+6.6–10.9 percentage points under FGSM and +17.1–26.3 percentage points under GA). Higher normalized AURC, stronger stability across FGSM budget and dimension sweeps, and lower attack-time decision margins all support the same ordering, while both Norm and Denorm fusion outperform monolithic embeddings and Denorm is often more stable at larger GA budgets. These findings support HDC-based classical-neural fusion as a practical direction for robustness under the evaluated threat models; future work will extend the evaluation to stronger iterative and adaptive attacks such as PGD and AutoAttack, measure runtime and memory overhead, and explore differentiable classical feature extraction for end-to-end tuning.

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